

Predicting Corporate Bond YTM Changes Using Machine Learning

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1. Objective

The objective of this project is to predict corporate bond yield to maturity (YTM) changes using firm-level fundamentals, market-based risk factors, and macroeconomic indicators.

By applying various machine learning regression models such as ElasticNet, Boosting and Multi-layer Perceptron, we aim to improve prediction accuracy compared to traditional linear regression models.

In addition, we used the change direction of YTM as a classification target since there are many outliers in the YTM change values that may affect regression performance. We applied classification models such as Logistic Regression, ElasticNet Classifier, and Boosting Classifier to predict the direction of YTM changes.

2. Methodology

2.1 Data Description

We will utilize data from Wharton Research Data Services (WRDS) to construct our dataset. The data frequency is monthly and data range is from July 2002 to February 2025.

2.1.1 Target Variable The target variable is the corporate bond yield to maturity (YTM) change, defined as the difference in YTM over a one-month horizon:

$$\Delta YTM_i = YTM_{i,t} - YTM_{i,t-1}$$

where $YTM_{i,t}$ is the yield to maturity of bond i at the end of month t .

For classification, the target variable is defined as:

$$YTM_Direction_i = \begin{cases} up, & \text{if } \Delta YTM_i > 1e-6 \\ neutral, & \text{if } \Delta YTM_i \approx 0 \\ down, & \text{if } \Delta YTM_i < -1e-6 \end{cases}$$

The Data is from the “WRDS Bond Returns” database, which provides monthly bond yields and other bond characteristics, sourced from TRACE and Mergent FISD. The dataset has been cleaned and more suitable for analysis.

We only consider senior bonds and exclude defaulted bonds. Additionally, the yields exceeding $[-1, 1]$ are treated as outliers and removed from the dataset.

Total number of observations meeting the criteria is approximately 1760852, covering 74191 unique bonds and 3187 unique firms over the sample period.

2.1.2 Features

1. Key Columns

- Bond Identifier: CUSIP
- Firm Identifier: PERMNO
- Date: End of Month

We have checked the missingness and uniqueness of these key columns.

2. Bond-level features

- Time to Maturity (tmt)
- Coupon Rate (coupon)
- t_spread: the weighted average bid-ask spread of the bond which measures liquidity.
- return_eom: bond return of last month.
- Credit Ratings: One-hot encoded variables for ratings from AAA to D (rating_A, rating_AA, ..., rating_D). The ratings are weighted averages ratings from WRDS Bond Returns database rather than S&P, Moody's, or Fitch alone.
- Rating Change Indicators: Binary variables indicating whether there was an upgrade or downgrade in the bond's credit rating in the past month (upgrade, downgrade).

Here, we used lagged values of features except time to maturity and coupon rate to avoid look-ahead bias.

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
tmt	3.56e+6	6.35	8.67	0.00	1.61	3.54	6.89	102.13
coupon	3.56e+6	0.03	0.03	0.00	0.00	0.03	0.06	0.34
t_spread	1.79e+6	0.01	0.01	0.00	0.00	0.00	0.01	2.00
yield	3.39e+6	0.01	0.13	-1.00	0.00	0.03	0.05	1.00
ret_eom	3.42e+6	0.01	1.18	-1.00	0.00	0.00	0.01	2090.0
rating_A	3.42e+6	0.20	0.40	0.00	0.00	0.00	0.00	1.00
rating_AA	3.42e+6	0.04	0.21	0.00	0.00	0.00	0.00	1.00
rating_AAA	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_B	3.42e+6	0.03	0.16	0.00	0.00	0.00	0.00	1.00
rating_BB	3.42e+6	0.04	0.20	0.00	0.00	0.00	0.00	1.00
rating_BBB	3.42e+6	0.23	0.42	0.00	0.00	0.00	0.00	1.00
rating_C	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
rating_CC	3.42e+6	0.00	0.03	0.00	0.00	0.00	0.00	1.00

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
rating_CCC	3.42e+6	0.01	0.09	0.00	0.00	0.00	0.00	1.00
rating_D	3.42e+6	0.00	0.02	0.00	0.00	0.00	0.00	1.00
upgrade	3.42e+6	0.00	0.04	0.00	0.00	0.00	0.00	1.00
downgrade	3.42e+6	0.00	0.05	0.00	0.00	0.00	0.00	1.00

3. Firm-level features

We incorporate firm-level information at both the daily and quarterly frequency.

- **Daily Stock Price and Returns:** Includes the daily closing stock price (PRC), daily stock return (RET), and trading volume (VOL). These variables capture short-term market performance and trading activity.
- **Daily Trading Liquidity Measures:** Includes the daily bid and ask quotes (BIDLO, ASKHI) and the number of trades (NUMTRD) as indicators of market liquidity.
- **Quarterly Accounting Fundamentals:** Includes total assets (atq), total liabilities (ltq), common equity (ceqq), shares outstanding (cshoq), and quarterly net income (niq). These variables summarize the firm’s financial position and profitability at the quarterly reporting date.
- **Firm Identification:** The PERMNO identifier is present in both daily stock data and quarterly fundamental data, enabling direct alignment of firm-level information across time.

4. Macroeconomic features (yuxi)

2.2 Feature Construction (Yutung)

To enhance bond yield prediction, we construct a comprehensive set of features from firm-level, market-level, and macroeconomic information. All features are aligned to month-end and lagged by one month to prevent look-ahead bias, reflecting the information available to investors at the time of prediction.

1. **Firm-Level Stock Market Features (Monthly Aggregated)** Derived from CRSP daily data and aggregated to monthly frequency: